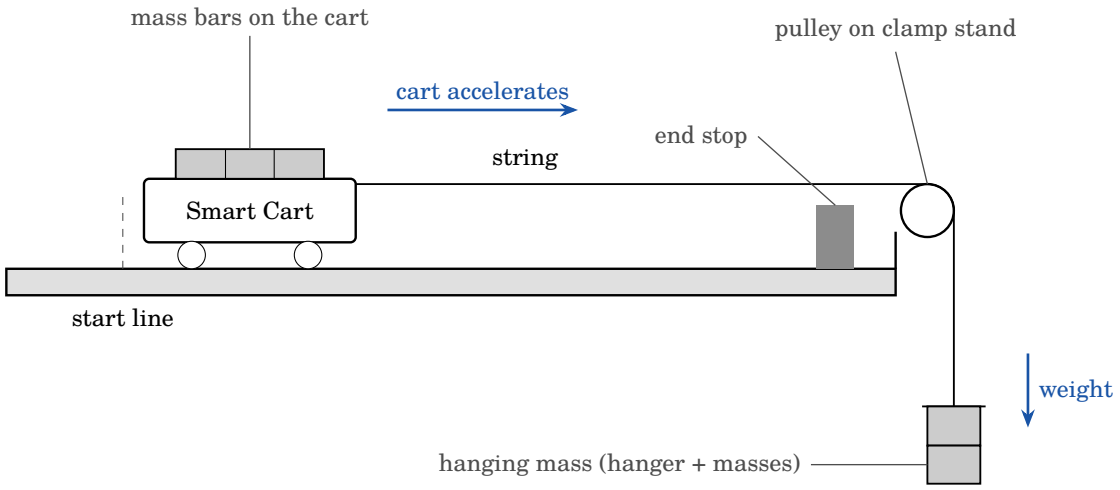
Name: _____

Class: _____

Date: _____

Mr. Rawson • GCSE Physics

RP19: Force, Mass and Acceleration



OPEN SPARKVUE ON THE LAPTOP

sparkvue.pasco.com

Set up, once

1. Level the track. Clamp the pulley at the table edge on its stand.
2. Open **sparkvue.pasco.com** on the laptop, connect the Smart Cart, and display the **velocity–time** graph.
3. Put the cart on the balance and write its mass here: _____ kg. Each mass bar is marked 250 g.
4. Tie the string to the cart, run it over the pulley and tie the hanger to the other end.
5. Pull the cart back to the start line, press record, release, and catch the cart.
6. Use **Linear Fit** on the straight, rising part of the velocity–time line while the hanger is falling. The **slope** m of that fit is the **acceleration** in m/s^2.

Name: _____ **Date:** _____

Part A: Constant Mass, Changing Force

Method

1. Put the **two 250 g bars** on the cart.
2. Take **five 50 g masses**. Hang **one** on the string and ride the other **four** on the cart.
3. Run 1: take the acceleration as in steps 5 and 6 of the set-up, and write it in the table.
4. For each new run, **move one 50 g mass from the cart to the hanger**. The hanging mass goes up by 50 g and the cart mass goes down by 50 g, so the two always add up to the same total.
5. Take the acceleration for Runs 2 to 5 the same way, and complete the other columns of the table.

Cart mass is everything on the cart side: the cart, the two bars and the 50 g masses still riding on it. **Total mass** is the cart mass plus the hanging mass.

Run	**Hanging mass / g**	**Cart mass / g**	**Total mass / g**	**Force F / N**	**Acceleration / m/s^2**	$a \div F$
1	50					
2	100					
3	150					
4	200					
5	250					

Check it. Cart mass plus hanging mass gives the total, and the total is the same number on every row. Work out the force in newtons: grams $\div$ 1000, then $\times$ 9.8. The last column, $a \div F$, should come out at about the same number on every row. That number is $1 \div M$, with M in kg, so it confirms the total mass stayed constant.

Name: _____ **Date:** _____

Part B: Constant Force, Changing Mass

Method

1. Keep **50 g hanging** for every run, so the force stays at $0.050 \times 9.8 = 0.49$ N.
2. Run 1 uses the **cart empty**, with no bars on it. Take the acceleration as in steps 5 and 6 of the set-up, and write it in the table.
3. For each new run, add **one more 250 g bar** to the cart. Each addition raises the total mass that the 0.49 N has to accelerate.
4. Take the acceleration for Runs 2 to 5 the same way, and complete the other columns of the table.

Cart mass is the cart plus the bars on it. The column assumes a 250 g cart, so write your own balance reading over it if your cart differs. **Total mass** is the cart mass plus the hanging mass.

Run	**Hanging mass / g**	**Cart mass / g**	**Total mass / g**	**Force F / N**	**Acceleration / m/s^2**	$a \times M$ **/ N**
1	50	250				
2	50	500				
3	50	750				
4	50	1000				
5	50	1250				

Check it. Cart mass plus hanging mass gives the total. Multiply each acceleration by the total mass in kg (the Total column $\div$ 1000). The last column, $a \times M$, is the force $F = m \times a$, so it should land close to the Force column, 0.49 N, on every run.

Name: _____ **Date:** _____

Graphs

All graphs are on the attached spreadsheet.

Add a trendline and its equation to each graph. Follow the clip for your part at

rawsonvault.com/physics/aqa-gcse/p5-forces/rp19

Part A: **Add a trendline: Part A, Constant Mass, Changing Force**
Part B: **Add a trendline to a curve: Part B, Constant Force, Changing Mass**

Analysis

Analysing a graph

Correlation describes how closely your points follow the trendline. **Perfect** correlation means every point sits exactly on the line, **strong** means every point sits close to the line, and **weak** means the points are spread well away from it.

An **anomaly** is a point that sits clearly away from the pattern that the other points make.

$$\textbf{Percent error} = \frac{\text{trendline value} - \text{theoretical value}}{\text{theoretical value}} \times 100$$

Below 5% error	**accurate**, and the data supports the theory
5% to 10% error	**somewhat accurate**
Above 10% error	the data is too far from the theory to support the formula

Part A: Constant Mass, Changing Force

1) **Analyse** Graph A, acceleration against force. State the **type of relationship** between acceleration and force, the **strength of the correlation** (perfect, strong or weak) with the reason you can see on your graph, and whether there are any **anomalies**. Name the run number of any point that sits clearly away from the trendline.

2) **Explain** how your trendline supports Newton's second law, $\Sigma F = m \cdot a$.

Your trendline from Graph A

Trendline equation: $a =$ _____ $F +$ _____

Theoretical gradient, $1 \div M$ (with M in kg): _____

Percent error of the gradient: _____

In your answer, say what the type of relationship means mathematically and how it matches $\Sigma F = m \cdot a$. Then use your percent error to say how closely your trendline matches the theory.

Part B: Constant Force, Changing Mass

1) **Analyse** Graph B, acceleration against total mass. State the **type of relationship** between acceleration and total mass, the **strength of the correlation** (perfect, strong or weak) with the reason you can see on your graph, and whether there are any **anomalies**. Name the run number of any point that sits clearly away from the trendline.

2) **Explain** how your trendline supports Newton's second law, $\Sigma F = m \cdot a$.

Your trendline from Graph B

Trendline equation: $a =$ _____ M to the power of _____

Theoretical k, the weight of the hanging mass in N: _____

Percent error of k: _____ Theoretical power of M: -1

In your answer, say what the type of relationship means mathematically and how it matches $\Sigma F = m \cdot a$. Then use your percent error to say how closely your trendline matches the theory.